Jazim Syed ([Linkedin])

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Education

Toronto Met University (*Formerly Ryerson)*, Biomedical Engineering (B. Eng) EXP. JUNE 2026

Relevant Courses: Microprocessors, Material Science, Electrical Systems, Biology, Control Systems

Professional Experience

Personal Support Worker, *Extendicare Long Term Care Home* AUGUST 2023 - PRESENT

- Supported nurses in a professional healthcare setting - to create a comfortable and healthy environment for 50+ residents.
- Enforced a plan to navigate Covid for both nurses, residents and 75+ visitors, allowing only individuals who tested negative with no symptoms into the building.
- Recorded and Analyzed resident behaviour and worked with nurses to assign required medications and additional support to residents.

Personal Projects (All projects can be viewed through: [Jazim's Github])

Developed an Arduino Based Breathalyzer with MQ-3 Sensor 2025

- Designed and built a low-cost breathalyzer prototype using an Arduino Uno Microcontroller and MQ-3 alcohol sensor, achieving a calibration value of 0.96 for alcohol vapor concentration detection.
- Incorporated a real-time alcohol monitoring system integrating user-specific factors (body weight, sex, time since last drink) to enhance BAC estimation precision.
- Optimized system costs to approximately $30 CAD total, delivering a 70% cheaper alternative to commercial breathalyzers without significant loss in reliability

Temperature Sensor Interfacing and Peripheral Control 2024

- Interfaced a TMP36 analog temperature sensor with a PIC24 microcontroller using Proteus 8.11, achieving real-time temperature monitoring with an ADC resolution of 0.488℃ per step
- Configured 10-bit ADC, UART communication (19200 baud rate), and Timer 2 to enable accurate sensor data acquisition and real-time display on the Virtual Terminal.
- Programmed dynamic control of a DC fan based on temperature thresholds, with fan direction toggling at 30℃ to stimulate biomedical thermal regulation systems.

Post-Stroke Rehabilitation Exoskeleton 2023

- Developed a 3D model of a rehabilitation exoskeleton for post-stroke patients using Autodesk Inventor. Theorized the required fabrics and materials to build the exoskeleton. Incorporated the use of a PCB board for the circuit design.
- Overcame challenges of integrating a power system with no external charging through the use of a renewable energy source.

Extracurricular Experience

VP Finance & Sponsorship Associate, Biomedical Engineering Course Union (BECU) - TMU 2022- PRESENT

- Planned budgets for the year (2000 CAD/yr)
- Planned and acquired funding through SOTI & Sanofi for the 2023 Biomedical Engineering Conference
- Built and kept track of all expenses and earned money through spreadsheets
- Engaged in communication with industry professionals and successfully secured their participation in BECU events.

Skills

- Healthcare Systems Familiarity
- Professional Communication
- Microsoft Office
- Event Planning
- Material Management
- Electrical Schematics

Software: C, C++, Python, Matlab, Solidworks, Quartus II, NI Multisim, Proteus 8.11, MPLAB X IDE